$$3PbO + 2NH_3 \longrightarrow 3Pb + N_2 + 3H_2O$$

**Reaction with acids:** When treated with acids it forms ammonium salts. This reaction shows that the affinity of ammonia for proton is greater than that of water.

**Reaction with chlorine and chlorides:** Ammonia reacts with chlorine and chlorides to give ammonium chloride as a final product. The reactions are different under different conditions as given below.

With excess ammonia

$$2\ NH_3 + 3\ Cl_2 \longrightarrow N_2 + 6\ HCl$$

$$6\ HCl + 6\ NH_3 \longrightarrow 6\ NH_4Cl$$

With excess of chlorine ammonia reacts to give nitrogen trichloride, an explosive substance.

$$2NH_3 + 6Cl_2 \longrightarrow 2NCl_3 + 6\ HCl$$

$$2NH_3(g) + HCl(g) \longrightarrow NH_4Cl\ (s)$$



Reaction of ammonia with HCL

**Formation of amides and nitrides:** With strong electro positive metals such as sodium, ammonia forms amides while it forms nitrides with metals like magnesium.

$$2Na + 2NH_3 \longrightarrow 2NaNH_2 + H_2$$

$$3Mg + 2NH_3 \longrightarrow Mg_3N_2 + 3H_2$$

**With metallic salts:** Ammonia reacts with metallic salts to give metal hydroxides (in case of Fe) or forming complexes (in case Cu)

$$Fe^{3+} + 3NH_4^+ \xrightarrow{3\ OH^-} Fe(OH)_3 + 3NH_4^+$$

$$Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$$

Tetraamminecopper(II)ion
(a coordination complex)

**Formation of amines:** Ammonia forms ammonated compounds by ion dipole attraction. Eg. $[CaCl_2.8NH_3]$. In this, the negative ends of ammonia dipole is attracted to $Ca^{2+}$ ion.

It can also act as a ligand and form coordination compounds such as $[Co(NH_3)_6]^{3+}$, $[Ag(NH_3)_2]^+$. For example when excess ammonia is added to aqueous solution copper sulphate a deep blue colour compound $[Cu(NH_3)_4]^{2+}$ is formed.

## Structure of ammonia

Ammonia molecule is pyramidal in shape N-H bond distance is 1.016 Å and H-H bond distance is 1.645 Å with a bond angle 107 °. The structure of ammonia may be regarded as a tetrahedral with

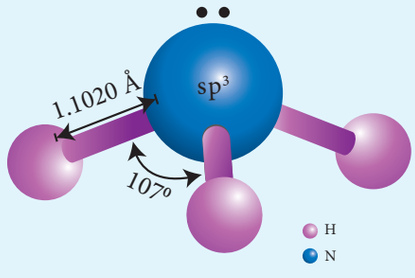



Figure 3.1 Structure of ammonia